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EXPERIMENT-1

OBJECTIVE:

Study of different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.

APPARATUS:

(Components): RJ-45 connector, Clipping Tool. Twisted pair Cable

PROCEDURE:

1. To do these practical following steps should be done: 1. Start by stripping off about 2 inches of the plastic jacket off the end of the cable. Be very careful at this point, as to not nick or cut into the wires, which are inside. Doing so could alter the characteristics of your cable, or even worse render is useless. Check the wires, one more time for nicks or cuts. If there are any, just whack the whole end off, and start over.
2. Spread the wires apart, but be sure to hold onto the base of the jacket with your other hand. You do not want the wires to become untwisted down inside the jacket. Category 5 cable must only have 1/2 of an inch of 'untwisted' wire at the end; otherwise it will be 'out of spec'. At this point, you obviously have ALOT more than 1/2 of an inch of un-twisted wire.
3. You have 2 end jacks, which must be installed on your cable. If you are using a pre-made cable, with one of the ends whacked off. you only have one end to install the crossed over end. Below are two diagrams, which show how you need to arrange the cables for each type of cable end. Decide at this point which end you are making and examine the associated picture below.

THEORY: **Types of Network Cables**

There are many types of cables available in the networking. Here we are going to discuss about six different types of cables.

1. Coaxial Cables: This cable was invented at late 1880's, which is used to connect the television sets to home antennas. This cable is used to transfer the information in 10 mbps. The cable is divided into thinnet and thicknet cables. These cables have a copper wire inside and insulation is covered on the top of the copper wire to provide protection to the cable. These cables are very difficult to install and maintain, because they are too big to carry and replace. The coaxial cable got its name by the word "coax". Nowadays coaxial cables are also used for dish TV where the setup box and the television is connected using the coaxial cable only. Some of the cable names are Media Bridge 50-foot Coaxial cable, Amazon basics CL2-Rated Coaxial cables, etc. See Figure 13.2



Figure13.2 Coaxial cables for connecting television sets

2. Twisted Pair Cables: It is type of cable with two or more insulated wires twisted together. It started with a speed of 10 mbps (10BASE-T cable is used). Then the cable was improved and the speed was higher and went to 100 mbps and the cable was renamed as 100BASE-TX. Then finally the cable was improved to 10 gbps and named as 10GBASE-T. This twisted cable has 8 wires which are twisted to ignore electromagnetic interference. The eight wires cannot be placed in a single unit as there can be difficulty in space, so it is twisted to make it one wire. There are two types of twisted pair cables, Unshielded Twisted Pair (UTP) and Shielded Twisted pair (STP). The UTP is used nowadays as modern cables for Internet and they are lower in cost and installation and maintenance is easy compared to the coaxial cables. STP is similar to UTP, but it is covered by an additional jacket to protect the wires from External interference. See Figure 13.3



Figure13.3 Twisted pair cables

3. Fiber Optics: This cable is different from the other two cables. The other two cables had an insulating material outside and the conducting material like copper inside. But in this cable it is strands of glass and pulse of light is used to send the information. They are mainly used in Wide Area Network (WAN). The WAN is a network that extends to very large distance to connect the computers. One example of WAN is Internet. These cables are placed in deep underground to avoid any damage to the cables. The optic cable uses light to transmit the information from one place to another. There are two types of fiber optic cables available, One is single-mode (100BaseBx) another one is Multimode (100BaseSX). Single-mode cables are used for long distance transmission and at a high cost whereas the multimode cables are used for short distance transmission at a very low cost. The optic cables are easy to maintain and install. See Figure 13.4



Figure 13.4 Fiber optic cable with 4 cores combined with one cable.

4. USB Cables: The Universal Serial Bus are used to connect keyboard, mouse and other peripheral devices. But there are some special network devices used to connect the Internet through the USB called dongles. The dongle is a small peripheral device which has a compatible of mobile broadband with a slim slot in it and connects the Internet and acts as a modem to the computer. See Figure 13.5

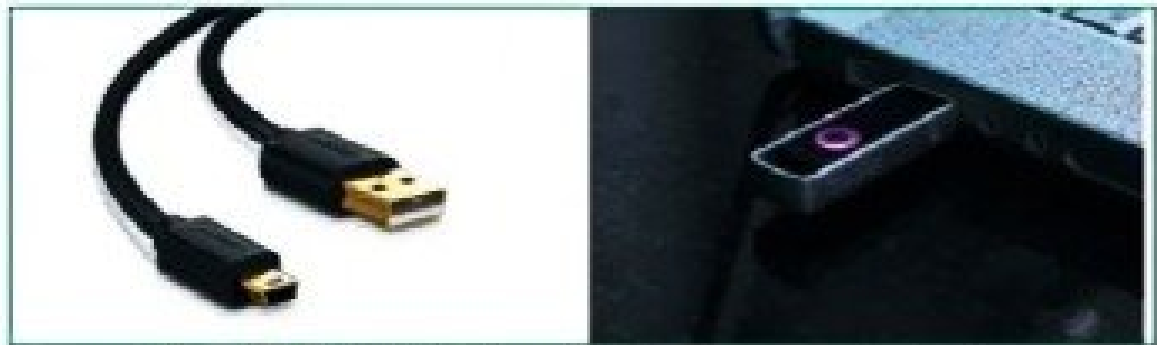


Figure 13.5 The USB Cables and USB dongle connected to a laptop.

The latest version of USB is USB 3.0 which has the data transfer rate 4.85 Gbps. But USB 2.0 has just 480 Mbps.

Micro USB is a miniaturized version of the USB used for connecting mobile devices such as smart phones, GPS devices and digital cameras.



5. Serial and Parallel cables: Earlier in 1980s to 1990s the Ethernet and the USB were not developed. Then the Serial and Parallel interface cables were used to connect the Internet to the system. They were sometime used for PC to PC networking. Before the USB emerged, the system will have both serial port and parallel port. The serial port will send 1 bit at one time whereas the parallel port will send 8 bit at one time. The parallel cables are used to connect to the printer and other disk drivers. RS232 is one type of serial cable, also known as Null modem cable. See Figure 13.6



Figure 13.6 The serial cable at the left and parallel cable at the right.

6. **Ethernet Cables:** Ethernet cable is the most common type of network cable mainly used for connecting the computers or devices at home or office. This cable connects wired devices within the local area network (LAN) for sharing the resources and accessing Internet.

The Crossover Ethernet cable is an example of the Null modem Cables. This cable is used to join two PCs or two network devices of the same type. This cable is the sophisticated Ethernet cable used to connect the Internet to the system. This cable works at a speed of 10 gbps and more. The Ethernet crossover cable is identical on both the ends. Nowadays Routers are being connected through the crossover cables to provide wireless network from the local network. See Figure 13.7



Figure 13.7 The Ethernet cables.

Just like the crossover cable, RS-232 cable is also used for interconnecting two computers without modem. So it is also a null modem cable. A cable interconnecting two devices directly is known as a null modem cable.



Diagram shows you how to prepare Cross wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Diagram shows you how to prepare straight through wired connection

Diagram shows you how to prepare straight through wired connection

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

EXPERIMENT-2

OBJECTIVE:

Study of Network Devices in Detail.

THEORY:

Network Devices

Networking hardware, also referred to as network equipment or computer networking devices, are electronic devices that allow important network devices on a computer network to communicate with one another.

Computers, printers, and other electronic devices can connect to a network via network devices. Many gadgets employ the same and different networks to send data quickly, safely, or accurately. There are both inter-network and intra-network network devices available. Specific components, including a network interface card or an RJ45 connection, were installed on the device, while others, such as a router or switch, are meaning of the art of the network. Let's take a deeper look at the meaning of network devices and many types of network equipment. This issue must be studied to gain a better understanding. So let's get started talking about network devices' meaning.

What are Network Devices?

A computer network's network devices allow various hardware components to connect. Physical devices, networking hardware, and network equipment are all terms used to describe these devices, also known as computer networking devices. Each network device in a computer network performs a certain function based on its capabilities, so it serves different objectives at various segments.

Important Network Devices

Hub

Connecting network cables to a connectivity device creates a network segment. A Hub is a name for this type of communication equipment. In most cases, hubs do not filter data; instead, they route incoming data frames or packets to any or all of the constituents. Nowadays, all-important network devices connected to PCs need a central switch or hub.

Repeater

The physical layer could be used to operate a repeater. This device's primary purpose is to duplicate the signal on a similar network before it becomes weak or broken. The critical thing to remember about these gadgets is that they don't boost the signal. They recreate the signal at its actual power whenever it becomes weak. A repeater is a gadget with two ports.

Bridge

In a computer network, a bridge connects two or more network segments. A bridge's primary function in network design is storing and distributing frames between different components. Transport frames are connected using (Media Access Control) technology.

These can be used to link two physical local area networks together to form a bigger logical local area network. Bridges act on the OSI model's data connection and physical layers to divide large and tiny networks by managing data flow between networks. Bridges have now been substituted with switches in recent years to provide more functionality.

NAS (Network-attached storage)

A network-attached storage (NAS) server is a server specialised in filing storage. A NAS provides a central storage point for a LAN that can be utilised for shared file access and storing user data backups. NAS devices are a cost-effective and straightforward approach to delivering network storage. As NASes begin to offer more advanced functionality suitable for small and mid-sized workplaces, the distinctions between a NAS device and a general-purpose server have been even more blurred in recent times.

Network Switch

A switch and a hub generally operate at the same LAN layer, but a switch is more intelligent than a hub. The hub transmits data, but the switch filters and forwards it. As a result, this is the most creative data packet handling method. When something is received from the switch's interface, a data packet can be filtered and forwarded to the indicated receiver's interface. As a result, a switch keeps track of system settings in a content addressable memory table, including memory. The FIB or the forwarding table are other names for this table.

Router

The Brouter is also known as a bridging router, and its primary function is to combine the functions of both a router and just a bridge. It can operate at either the network or data link layer. When it functions as a router, it routes packets across networks, and then when it functions as a bridge, it filters LAN traffic.

Gateway

A gateway works at the OSI model's session and transport layers. TCP/IP and other networking protocols are transformed through gateways. They're connected to two or more independent networks, each with its domain name service, routing algorithm, topology, and network administration, including policy procedures. Gateways execute all of the functions that routers do. A gateway is a router that also has the capability of converting traffic. As a result, a protocol converter is used to switch between network technologies.

Conclusion

A network device, such as a fax machine or printer, is a type of equipment that connects devices or computers to transfer resources or files. A network consists of two or more computer systems connected by a telecommunications system to talk and share resources. Companies would not be able to share resources and enhance efficiency as effectively if they did not have a network. I hope now you got all the necessary information about network devices meaning. For better understanding, you must read this topic thoroughly to know some of the important network devices.



Types of Network Devices

EXPERIMENT-3

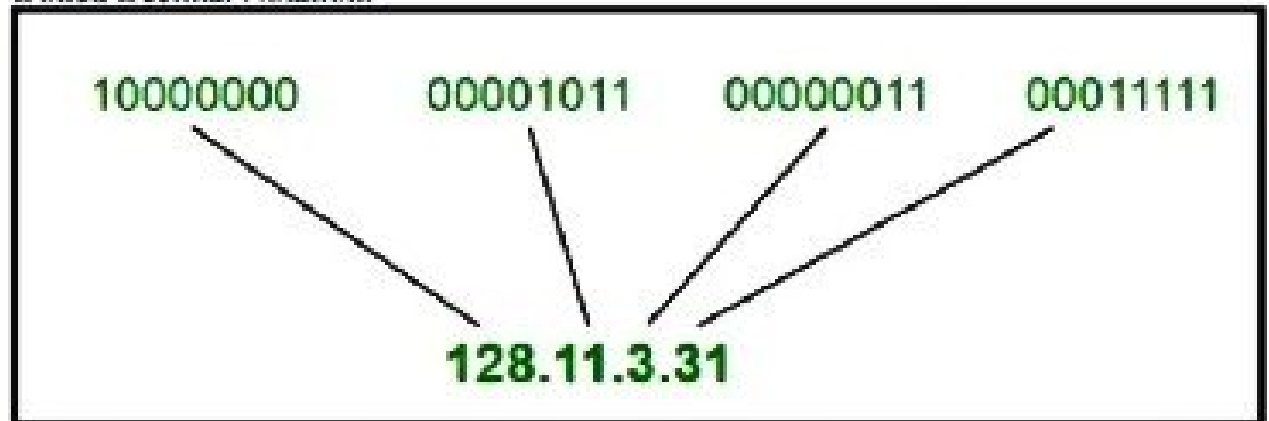
OBJECTIVE:

Study of network IP.

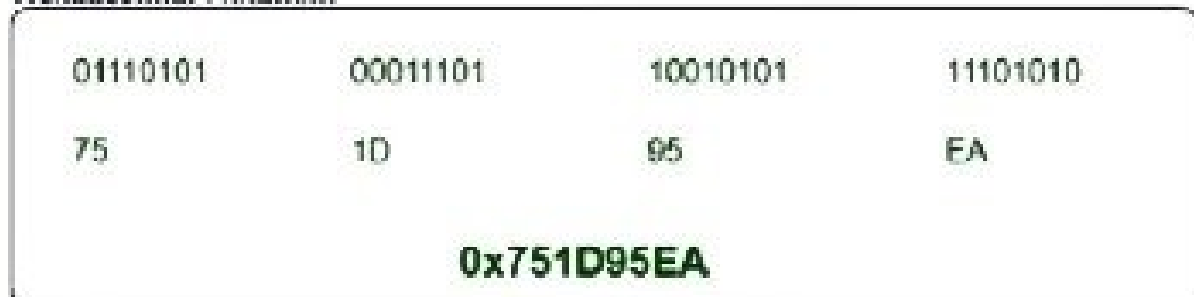
THEORY:

IP address is an address having information about how to reach a specific host, especially outside the LAN. An IP address is a 32 bit unique address having an address space of 2^{32} . Generally, there are two notations in which IP address is written, dotted decimal notation and hexadecimal notation.

Dotted Decimal Notation:



Hexadecimal Notation:



Some points to be noted about dotted decimal notation:

1. The value of any segment (byte) is between 0 and 255 (both included).
2. There are no zeroes preceding the value in any segment (054 is wrong, 54 is correct).

Classful Addressing

The 32 bit IP address is divided into five sub-classes. These are:

- Class A
- Class B
- Class C
- Class D
- Class E

Each of these classes has a valid range of IP addresses. Classes D and E are reserved for multicast and experimental purposes respectively. The order of bits in the first octet determine the classes of IP address.

IPv4 address is divided into two parts:

- Network ID
- Host ID

The class of IP address is used to determine the bits used for network ID and host ID and the number of total networks and hosts possible in that particular class. Each ISP or network administrator assigns IP address to each device that is connected to its network.

Note: IP addresses are globally managed by Internet Assigned Numbers Authority(IANA) and regional Internet registries(RIR).

Note: While finding the total number of host IP addresses, 2 IP addresses are not counted and are therefore, decreased from the total count because the first IP address of any network is the network number and whereas the last IP address is reserved for broadcast IP.

Class A:

IP address belonging to class A are assigned to the networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.

The higher order bit of the first octet in class A is always set to 0. The remaining 7 bits in first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default subnet mask for class A is 255.x.x.x. Therefore, class A has a total of:

- $2^7 - 2 = 126$ network ID (Here 2 address is subtracted because 0.0.0.0 and 127.x.y.z are special address.)
- $2^{24} - 2 = 16,777,214$ host ID

IP addresses belonging to class A ranges from 1.x.x.x – 126.x.x.x

7 Bit		24 Bit
0	Network	Host

Class A

Class B:

IP address belonging to class B are assigned to the networks that ranges from medium-sized to large-sized networks.

- The network ID is 16 bits long.
- The host ID is 16 bits long.

The higher order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine network ID. The 16 bits of host ID is used to determine the host in any network. The default sub-net mask for class B is 255.255.x.x. Class B has a total of:

- $2^{14} - 16384$ network address
- $2^{16} - 2 - 65534$ host address

IP addresses belonging to class B ranges from 128.0.x.x – 191.255.x.x.



Class B

Class C:

IP address belonging to class C are assigned to small-sized networks.

- The network ID is 24 bits long.
- The host ID is 8 bits long.

The higher order bits of the first octet of IP addresses of class C are always set to 110. The remaining 21 bits are used to determine network ID. The 8 bits of host ID is used to determine the host in any network. The default sub-net mask for class C is 255.255.255.x. Class C has a total of:

- $2^{21} - 2097152$ network address
- $2^8 - 2 - 254$ host address

IP addresses belonging to class C ranges from 192.0.0.x – 223.255.255.x.



Class C

Class D:

IP address belonging to class D are reserved for multi-casting. The higher order bits of the first octet of IP addresses belonging to class D are always set to 1110. The remaining bits are for the address that interested hosts recognize.

Class D does not possess any sub-net mask. IP addresses belonging to class D range from 224.0.0.0 – 239.255.255.255.

28 Bit

1	1	1	0	Host
---	---	---	---	------

Class D

Class E:

IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E range from 240.0.0.0 – 255.255.255.254. This class doesn't have any sub-net mask. The higher order bits of first octet of class E are always set to 1111.

28 Bit

1	1	1	1	Host
---	---	---	---	------

Class E

Range of special IP addresses:

169.254.0.0 – 169.254.0.16 : Link local addresses

127.0.0.0 – 127.0.0.8 : Loop-back addresses

0.0.0.0 – 0.0.0.8 : used to communicate within the current network.

Rules for assigning Host ID:

Host ID's are used to identify a host within a network. The host ID are assigned based on the following rules:

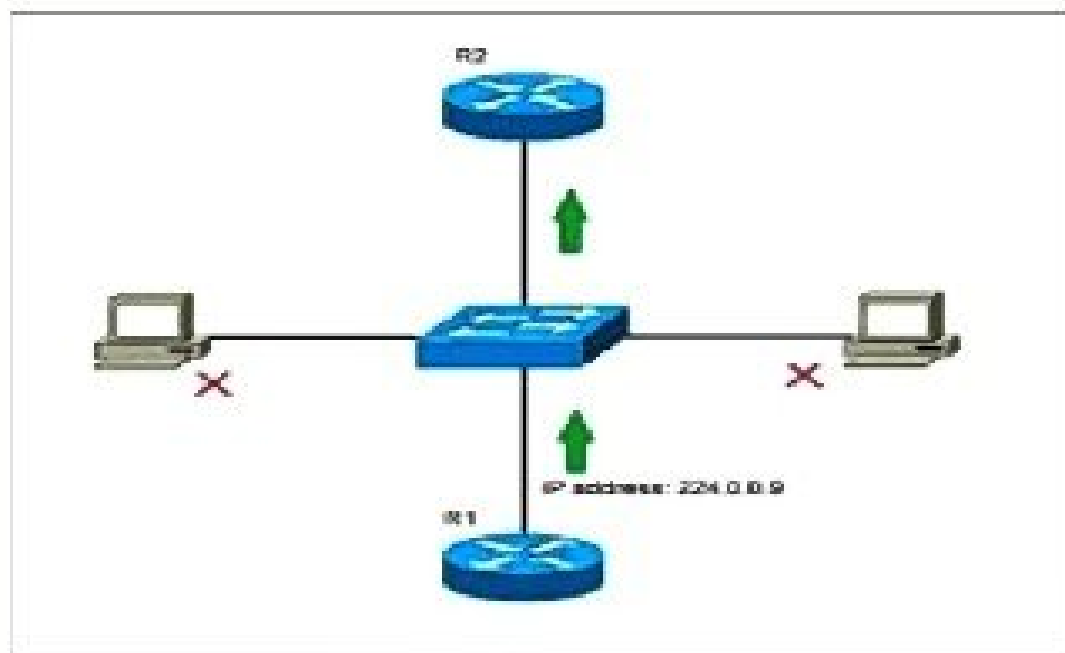
- Within any network, the host ID must be unique to that network.
- Host ID in which all bits are set to 0 cannot be assigned because this host ID is used to represent the network ID of the IP address.
- Host ID in which all bits are set to 1 cannot be assigned because this host ID is reserved as a broadcast address to send packets to all the hosts present on that particular network.
-

Rules for assigning Network ID:

Hosts that are located on the same physical network are identified by the network ID, as all host on the same physical network is assigned the same network ID. The network ID is assigned based on the following rules:

- The network ID cannot start with 127 because 127 belongs to class A address and is reserved for internal loop-back functions.

- All bits of network ID set to 1 are reserved for use as an IP broadcast address and therefore, cannot be used.
- All bits of network ID set to 0 are used to denote a specific host on the local network and are not routed and therefore, aren't used.



EXPERIMENT-4

OBJECTIVE:

Connect the computers in Local Area Network.

PROCEDURE:

How to Connect to a Computer on a Local Area Network

Access computers on your Local Area Network and provide support and maintenance without end-user interaction.

Before you begin: The agent's Technician Group must have **Connect On LAN** permission.

Restriction: The Connect On LAN feature is not available for Technician Console for Mac.

1. On the Session Toolbar, click the **Computers** icon.



Result: The Computers panel is displayed.

2. On the Computers list, click the **Connect On LAN** tab to see a list of accessible computers.
3. Filter computers by name or IP address.

Tip: You can search and connect at any time. You do not need to wait for the entire list of computers to load.

4. Select the computer you want to access and click **Connect**.

Important: If you do not have Administrator rights, you are prompted to enter the login credentials of the computer you want to access.

*Result: The session appears in the Technician Console in **Connecting** status. The applet launches on the remote computer and the session starts. The customer does not have to grant access to the computer, so you can manage it without customer interaction.*

- Required Windows settings:
 - File and Printer sharing must be allowed on the customer's firewall.

- **Network access:** Sharing and security model for local accounts must be set to **Classic** - local users authenticate as themselves on the customer's computer (**Control Panel > [System and Security In Win7] > Administrative Tools > Local Security Policy > Local Policies > Security Options > Network access: Sharing and security model for local accounts**).
- Both the technician and customer computer must use a compatible NTLM authentication method: **Control Panel > [System and Security In Win7] > Administrative Tools > Local Security Policy > Local Policies > Security Options > Network security: LAN Manager authentication level** (for example, set both to Send NTLMv2 response only).
- If the Remote Registry service is enabled on the remote computer, the Technician Console will use that. (Go to **Control Panel > Administrative Tools > Services**, right-click **Remote Registry** and click **Properties**. Set **Startup type** to **Automatic** or **Manual**.) If the Remote Registry service is disabled, the Technician Console can also use Windows Management Instrumentation command line utility (WMIC) as a fall-back solution. WMIC is more secure, but slower.
- **NetBIOS** uses ports 135, 137, 138, 139
- Windows uses NTLM authentication to acquire administrator rights. This authentication uses randomly allocated TCP ports: 1024-65535, or 49152-65535 on Vista and Windows Server 2008. Communication through these ports must be allowed by firewalls.
- Computer Detection modes

Technician Console version 7.12.3351 or lower

Previous versions of the Technician Console used the command line tool `net.exe` to detect computers on a network. The latest versions of Windows disable the SMBv1 protocol, on which the tool is based. To restore the original functionality, the customer needs to re-enable SMBv1 on the remote computers.

Technician Console version 7.12.3352 or above

If the Active Directory Lightweight Directory Services Windows feature is enabled on the technician's computer, the Technician Console can read the computer list from Active Directory.

THEORY:

What Is a Local Area Network?

The LAN is a private network that allows devices on it to connect to each other and share data. It covers a specific area through at least one router. The exact size of that area may

vary from a house to an office to an entire office building. All devices connected to the LAN share a network connection.

There are a couple standard pieces of hardware found in any LAN:

- **Modem:** This enables an Internet connection from your ISP.
- **Router:** This sets up the local network and creates WiFi from the modem's Internet connection. Consider it the border between your private network and public cyberspace—aka where the firewall lives. Devices within the LAN can connect to each other and share data (hence the name: routing data) without necessarily needing an Internet connection. A simple LAN setup, like what you would find at home. (Image via Cloudflare)

Fun fact: LANs predate the Internet because the primary function is to connect devices for data sharing—technically no Internet connection is required.

A household LAN likely needs a single router. Larger networks can expand range through additional routers and switches to connect as many devices as necessary, even bringing in a dedicated server. It's the same functionality across the board, just scaled up as necessary. Business-grade hardware usually has beefed-up security protocols to better protect the larger network.

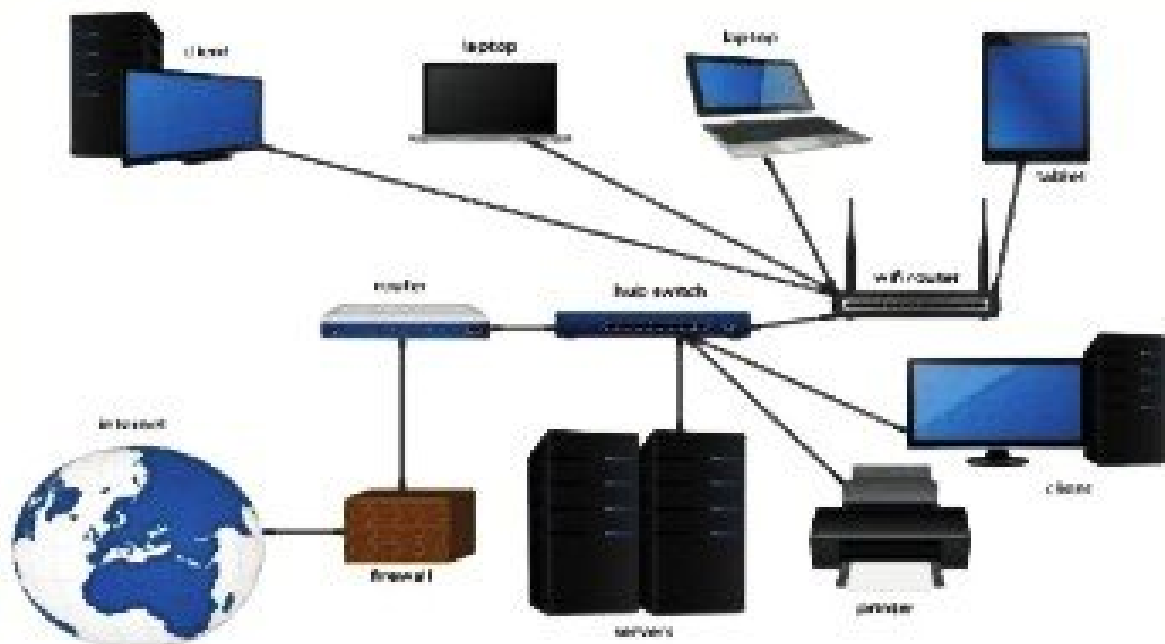
Example of a Local Area Network

Take an office building, for example. You can send files from your computer to another device like a printer or fax machine without having to directly connect them.

There are a few expansions on LANs:

- **Wide area network (WAN):** Several LANs connected over a larger area with no restrictions on how far that area extends.
- **Metropolitan area network:** A group of connected LANs in a single metropolitan area, like a city.
- **Campus area network (CAN):** A group of connected LANs across a school campus.

For example, a school administrative building will have its own LAN for that building alone. A separate classroom building will also have its own LAN. But as part of the CAN, the two LANs can connect to each other to essentially be on the same network.



A more robust LAN, like what you might find at a corporate office.

How to Connect Computers via LAN

There are two ways to connect a device to the LAN: Ethernet and WiFi.

1. Ethernet, or a wired connection, requires an Ethernet cable to physically plug in to the router or switch. If a device can't connect via WiFi or the signal is spotty, a wired connection will provide solid access to the network. If this sounds familiar, it should! VoIP phones are connected and powered over Ethernet.
2. WiFi, or a wireless connection, picks up the signal from the router without wires getting in the way.

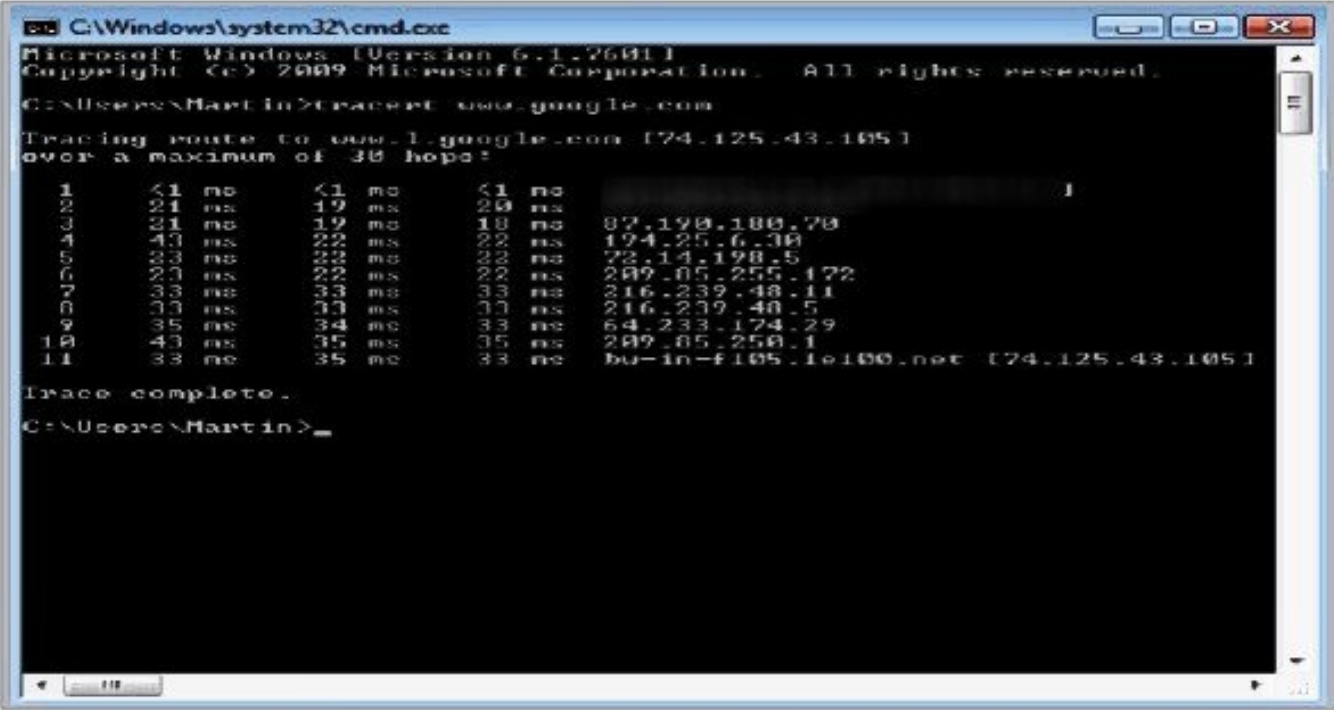
Setting up the LAN is the same as setting up your WiFi at home. You just plug in the modem and router, name your network (SSID), and set a password. Any wirelessly connected devices just need the password to connect.

EXPERIMENT-5

OBJECTIVE: Study of basic network command and Network configuration commands.

1. TRACERT COMMAND:- The tracert command is a Command Prompt command that's used to show several details about the path that a packet takes from the computer or device you're on to whatever destination you specify.

Tracert Command Option tracert -d :- This option prevents tracert from resolving IP addresses hostnames to , often resulting in much faster results



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Martin>tracert www.google.com

Tracing route to www.l.google.com [74.125.43.105]
over a maximum of 30 hops:
  0  <1 ms    <1 ms    <1 ms    1
  1  21 ms     19 ms     20 ms    1
  2  21 ms     19 ms     18 ms    87.198.188.78
  3  43 ms     22 ms     22 ms    194.25.6.38
  4  23 ms     22 ms     22 ms    72.14.198.5
  5  23 ms     22 ms     22 ms    209.85.255.172
  6  33 ms     33 ms     33 ms    216.239.48.11
  7  33 ms     33 ms     33 ms    216.239.48.5
  8  35 ms     34 ms     33 ms    64.233.174.29
  9  43 ms     35 ms     35 ms    209.85.258.1
 10  33 ms     35 ms     33 ms    bu-in-f105.1e100.net [74.125.43.105]

Trace complete.

C:\Users\Martin>
```

2. Ping Command:- Ping is a command-line utility, available on virtually any operating system with network connectivity, that acts as a test to see if a networked device is reachable. The ping command sends a request over the network to a specific device. A successful ping results in a response from the computer that was pinged back to the originating computer.

```

Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\User>ping webhostface.com

Pinging webhostface.com [108.178.8.52] with 32 bytes of data:
Reply from 108.178.8.52: bytes=32 time=134ms TTL=45
Reply from 108.178.8.52: bytes=32 time=134ms TTL=45
Reply from 108.178.8.52: bytes=32 time=135ms TTL=45
Reply from 108.178.8.52: bytes=32 time=134ms TTL=45

Ping statistics for 108.178.8.52:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 134ms, Maximum = 135ms, Average = 134ms

C:\Users\User>

```

3. **ARP Command :-** ARP stands for <Address Resolution Protocol> is a protocol for mapping an IP address to a physical MAC address on a local area network. Basically, ARP is a program used by a computer system to find another computer's MAC address based on its IP address. Communication between two computers on the same broadcast domain means a local area network. First, the client checks its ARP cache. ARP cache is a table of IP addresses with their corresponding MAC addresses.

```

Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\user>arp -a

Interface: 192.168.1.101 --- Bcb
Internet Address      Physical Address      Type
192.168.1.1           c8-3a-35-20-e6-e8     dynamic
192.168.1.50          00-4b-2b-5b-dc-a0     dynamic
192.168.1.102         00-00-7d-f6-62-e2     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
224.0.0.253           01-00-5e-00-00-fd     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\user>

```

Netstat Command :- The netstat command, meaning, is a command network statistics prompt command used to display detailed information very about how your computer is communicating with other computers or network devices.

Netstat Command Option

1. **-f netstat -f-** The switch will force the netstat command to display the Fully Qualified Domain Name (FQDN) for each foreign IP address when possible.

2. netstat -o :-A handy option for many troubleshooting tasks, the switch displays the process identifier (PID) -o associated with each displayed connection. See the example below for more about using

```
C:\Documents and Settings\Owner>netstat -an
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	127.0.0.1:1027	0.0.0.0:0	LISTENING
TCP	192.168.1.100:139	0.0.0.0:0	LISTENING
TCP	192.168.1.100:2559	207.88.172.236:80	CLOSE_WAIT
TCP	192.168.1.100:2916	204.14.90.25:21	CLOSE_WAIT
TCP	192.168.1.100:2923	69.89.109.55:80	TIME_WAIT
TCP	192.168.1.100:2924	204.245.162.25:80	ESTABLISHED
TCP	192.168.1.100:2929	66.150.98.119:80	ESTABLISHED
TCP	192.168.1.100:2930	204.245.162.27:80	ESTABLISHED
UDP	0.0.0.0:445	*:*	
UDP	0.0.0.0:500	*:*	
UDP	0.0.0.0:1030	*:*	
UDP	0.0.0.0:1040	*:*	
UDP	0.0.0.0:1155	*:*	
UDP	0.0.0.0:1175	*:*	
UDP	0.0.0.0:4500	*:*	
UDP	127.0.0.1:123	*:*	
UDP	127.0.0.1:1036	*:*	
UDP	127.0.0.1:1900	*:*	
UDP	127.0.0.1:2922	*:*	
UDP	192.168.1.100:123	*:*	
UDP	192.168.1.100:137	*:*	
UDP	192.168.1.100:138	*:*	
UDP	192.168.1.100:1400	*:*	

5. Nbtstat Command :- Nbtstat is a utility that displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP), which helps troubleshoot NetBIOS name resolution issues. Normally, name resolution is performed when NetBIOS over TCP/IP is functioning correctly. It does this through a local cache lookup, WINS or DNS server query or through LMHOSTS or hosts file lookup.

Netstat Command Option

1. **nbtstat -c :-** Lists NBT's cache of remote machine names and their IP addresses.

```
C:\WINDOWS\system32>nbtstat -c
```

Local Area Connection:
Node IpAddress: {192.168.1.101} Scope Id: {}

Name	Type	Host Address	Life (sec)
S-RMV-AS	UNIQUE	192.168.1.103	237
U-WMU-U2KPROP	UNIQUE	192.168.1.108	90
WIE	UNIQUE	192.168.1.108	17

```
C:\WINDOWS\system32>nbtstat -o
```

Local Area Connection:
Node IpAddress: {192.168.1.101} Scope Id: {}

Name	Type	Host Address	Life (sec)
S-RMV-AS	UNIQUE	192.168.1.103	238
U-WMU-U2KPROP	UNIQUE	192.168.1.108	62

```
C:\WINDOWS\system32>
```


6. IPConfig Command :- IPconfig is a console application designed to run from the Windows command prompt. This utility allows you to get the IP address information of a Windows computer . It also allows some control over your network adapters, IP addresses (DHCP-assigned specifically), even your DNS cache

```
Select Administration: Command Prompt

Microsoft Windows [Version 10.0.10240]
(c) 2015 Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : TechnologyRSS.local
    IPv4 Address. . . . . : 192.168.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : TechnologyRSS.local

Tunnel adapter isatap.TechnologyRSS.local:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : TechnologyRSS.local
```



7.Nslookup Command :- Nslookup is the name of a program that lets an Internet server administrator or any computer user enter a host name (for example, "whatiss.com") and find out the corresponding IP address or domain name system(DNS) record. The user can also enter a command for it to do a reverse DNS lookup and find the host name for an IP address that is specified.

```
File Edit View Terminal Help Help

nslookup -type=ns -domain=cloudns.net
Server: 10.0.0.1
Address: 10.0.0.1

Non-authoritative answer:
10.0.0.1 nscloudns.net = dns2.cloudns.net
10.0.0.1 nscloudns.net = dns4.cloudns.net
10.0.0.1 nscloudns.net = dns4.cloudns.net
10.0.0.1 nscloudns.net = dns2.cloudns.net
10.0.0.1 nscloudns.net = dns2.cloudns.net

Authoritative answers can be found from:
1. ■
```

EXPERIMENT-6

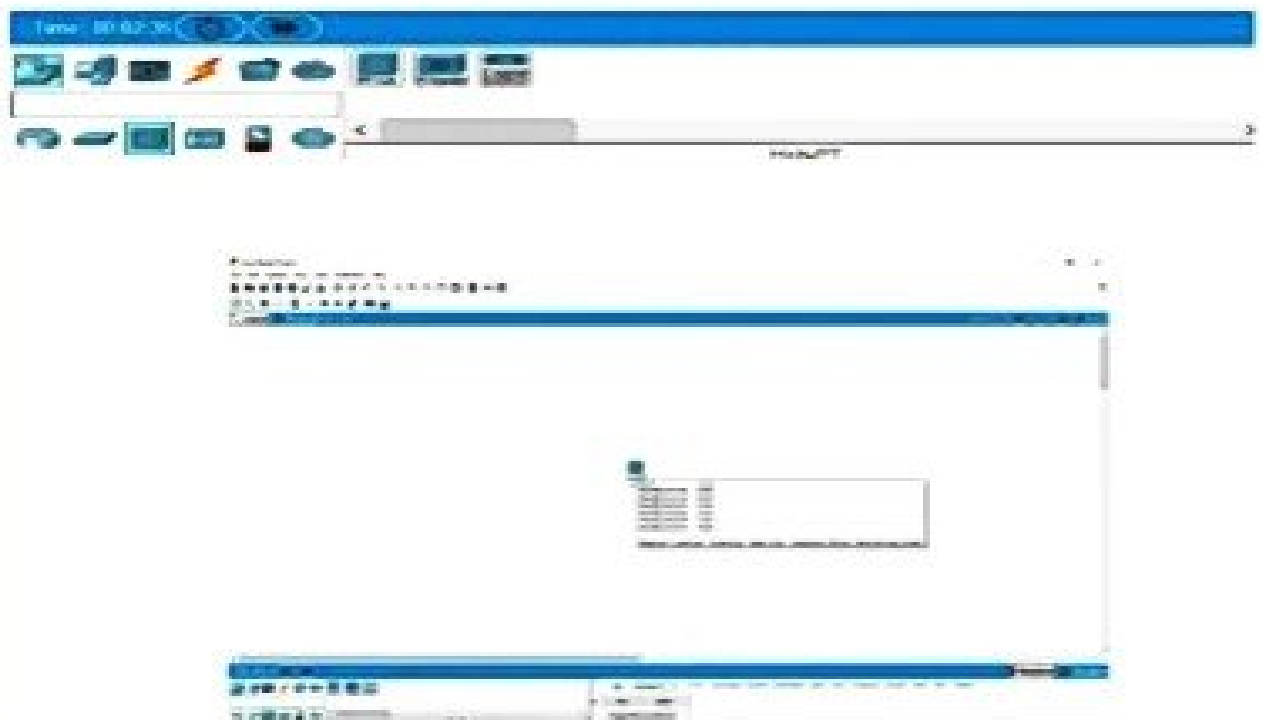
OBJECTIVE: Configure a network topology using packet tracer software.

THEORY : In this experiment we are building and implementing a simple star topology.

A star topology is a topology for a Local Area Network (LAN) in which all nodes are individually connected to a central connection point, like a hub or a switch. A star takes more cable than e.g. a bus, but the benefit is that if a cable fails, only one node will be brought down.

PROCEDURE

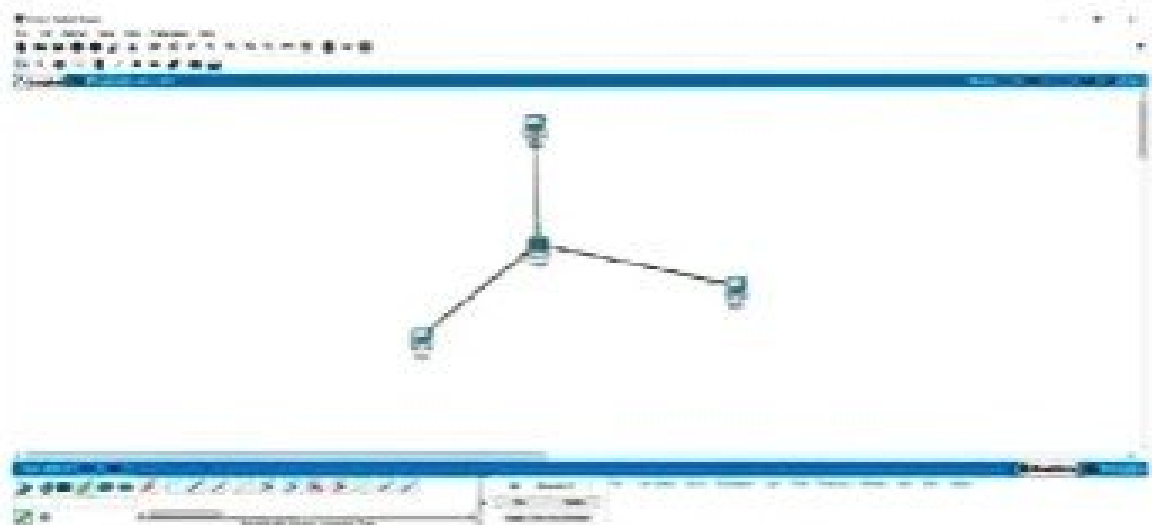
Step 1: Open Cisco Packet Tracer. Select Hub from Network Devices and place it on canvas.



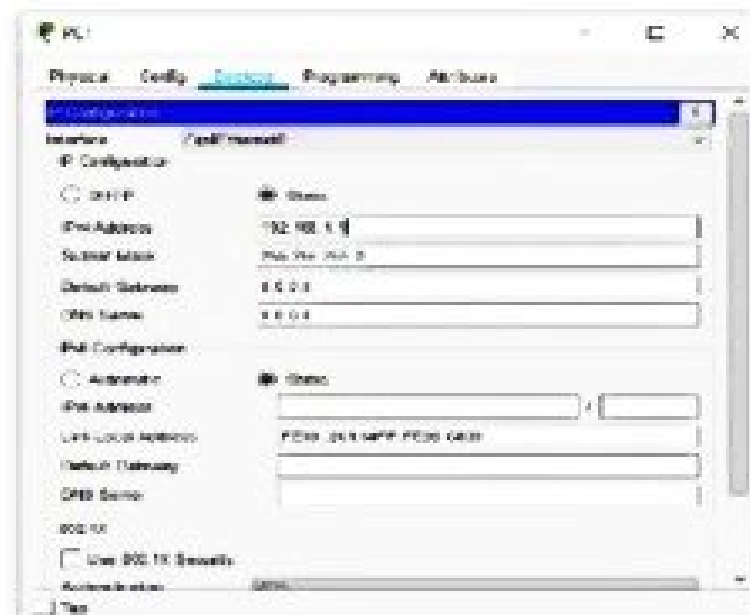
Step 2: Select few PCs or Laptops from End Devices and place them on canvas.



Step 3:- Select Automatic Connections from Connections and connect them as shown below.



Step 4: Click on a PC. Go to Desktop > IP Configuration and enter the IP as shown below



Step 5: After completing the setup, you now need to test your network. So click on Add simple PDU icon ,then select the any two PCs. After that Click on Simulation to see how your network works.



Conclusion:- From this experiment, we have learnt how to make a simple star network topology and implement the same. Similarly, we can also form any network topology like mesh, ring etc. using Cisco Packet Tracer.

EXPERIMENT-7

OBJECTIVE: Configure Network using Link State Vector Routing protocol.

PROCEDURE:

1. Develop a Topology shown in figure given below.
2. Configure all Routers
3. Configure all the nodes
4. Implement RIP protocols in Router to configure Network.

Router0 configuration:

CLI MODE:

Router>enable

Router# configure terminal

Router(config)#interface FastEthernet0/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit

Router(config)#interface Serial2/0

Router(config-if)#clock rate 64000

Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial2/0, changed state to down

Router(config-if)#ip address 10.0.0.2 255.0.0.0

Router(config-if)#ip address 10.10.0.2 255.0.0.0

Router(config-if)#%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#exit

Router# configure terminal

```
Router(config)#router rip
```

```
Router(config-router)#network 192.168.1.0
```

```
Router(config-router)#network 10.0.0.0
```

```
Router(config-router)#exit
```

GUI Mode:-

1. Configuring Network address

Physical Config CLI

GLOBAL	FastEthernet0/0
Settings	Port Status ☒ On
Algorithm Settings	Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
ROUTING	Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
Static	MAC Address 0050.0FC3.CB08
RIP	IP Configuration
INTERFACE	IP Address 192.168.1.1
FastEthernet0/0	Subnet Mask 255.255.255.0
FastEthernet1/0	
Serial2/0	
Serial3/0	
FastEthernet4/0	
FastEthernet5/0	

Tx Ring Limit 10

2. Configuring Router to Router connection

Physical Config CLI

GLOBAL	Serial2/0
Settings	Port Status ☒ On
Algorithm Settings	Duplex ☒ Full Duplex
ROUTING	Clock Rate 64000
Static	IP Configuration
RIP	IP Address 10.10.0.2
INTERFACE	Subnet Mask 255.0.0.0
FastEthernet0/0	
FastEthernet1/0	
Serial2/0	
Serial3/0	
FastEthernet4/0	
FastEthernet5/0	

Tx Ring Limit 10

3. Adding the IP address of the nodes connected to the router



Router1 Configuration:

```
Router>enable
```

```
Router# configure terminal
```

```
Router(config)#interface FastEthernet0/0
```

```
Router(config-if)#ip address 192.168.2.1 255.255.255.0
```

```
Router(config-if)#exit
```

```
Router(config)#interface FastEthernet0/0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

```
Router(config-if)#exit
```

```
Router(config)#interface Serial2/0
```

```
Router(config-if)#no shutdown
```

```
Router(config-if)#
```

```
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
```

```
ip address 10.10.0.3 255.0.0.0
```

```
Router(config-if)#
```

```
Router(config-if)#exit
```

```
Router(config)#interface Serial2/0
```

```
Router(config-if)#
```

```
Router(config-if)#exit
```

```
Router(config)#router rip
```

```
Router(config-router)#network 192.168.2.0
```

```
Router(config-router)#network 10.0.0.0
```

```
Router(config-router)#exit
```

1. Configuring Network Address

Physical Config CLI

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0004.9A60.6B40

IP Configuration

IP Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

2. Configuring Router to router connection

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate Not Set

IP Configuration

IP Address 10.10.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

PC0 Configuration:

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IP Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server

IPv6 Configuration

☐ DHCP ☐ Auto Config ☒ Static

IPv6 Address /

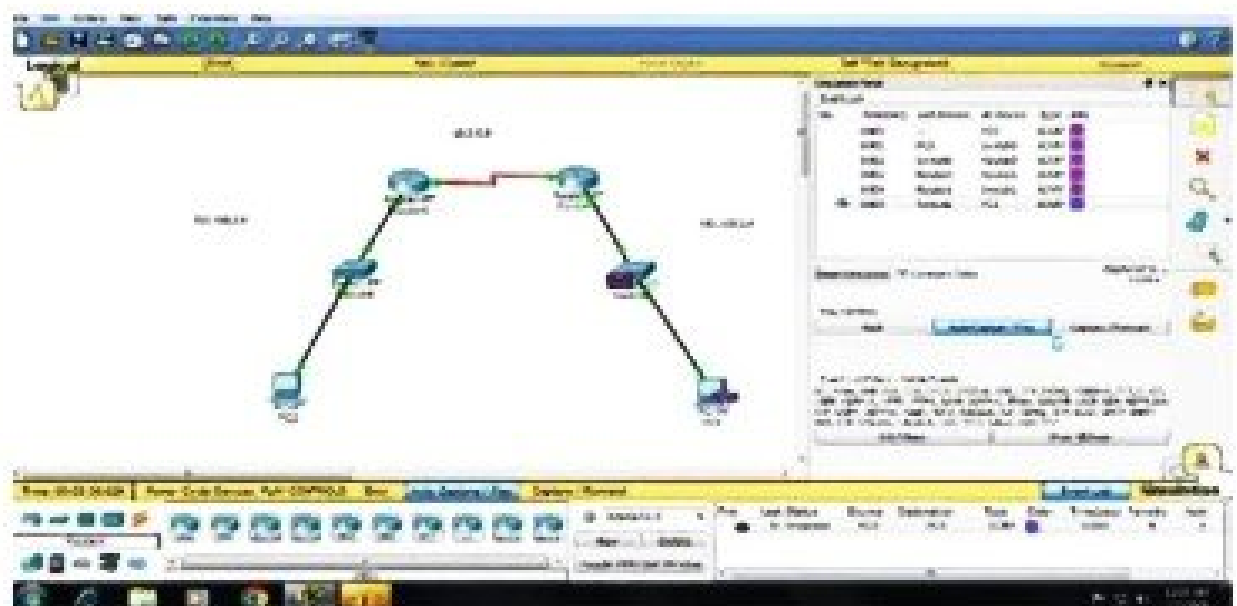
Link Local Address FE80::200:9CFF:FE73:7670

IPv6 Gateway

IPv6 DNS Server

PC1 Configuration:

IP Configuration		
IP Configuration		
☐ DHCP	☒ Static	
IP Address	192.168.2.2	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.2.1	
DNS Server	0.0.0.0	
IPv6 Configuration		
☐ DHCP	☐ Auto Config	☒ Static
IPv6 Address		
Link Local Address	FE80::20D:80FF:FECC:54E1	
IPv6 Gateway		
IPv6 DNS Server		



EXPERIMENT-8

OBJECTIVE: Configure Network using Link State Vector Routing protocol.

THEORY:

In link state routing, each router shares its knowledge of its neighborhood with every other router in the internet work.

(i) Knowledge about Neighborhood: Instead of sending its entire routing table a router sends info about its neighborhood only.

(ii) To all Routers: each router sends this information to every other router on the internet work not just to its neighbour. It does so by a process called flooding.

(iii) Information sharing when there is a change: Each router sends out information about the neighbors when there is change.

PROCEDURE:

1. The Dijkstra algorithm follows four steps to discover what is called the shortest path tree (routing table) for each router:
2. The algorithm begins to build the tree by identifying its roots. The root router's tree is the router itself.
3. The algorithm then attaches all nodes that can be reached from the root. The algorithm compares the tree's temporary arcs and identifies the arc with the lowest cumulative cost. This arc and the node to which it connects are now a permanent part of the shortest path tree.
4. The algorithm examines the database and identifies every node that can be reached from its chosen node. These nodes and their arcs are added temporarily to the tree.
5. The last two steps are repeated until every node in the network has become a permanent part of the tree.

ALGORITHM:

1. Create a simulator object
2. Define different colors for different data flows
3. Open a nam trace file and define finish procedure then close the trace file, and execute nam on trace file.
4. Create n number of nodes using for loop
5. Create duplex links between the nodes
6. Setup UDP Connection between n(0) and n(5)
7. Setup another UDP connection between n(1) and n(5)
8. Apply CBR Traffic over both UDP connections
9. Choose Link state routing protocol to transmit data from sender to receiver.
10. Schedule events and run the program.

PROGRAM:

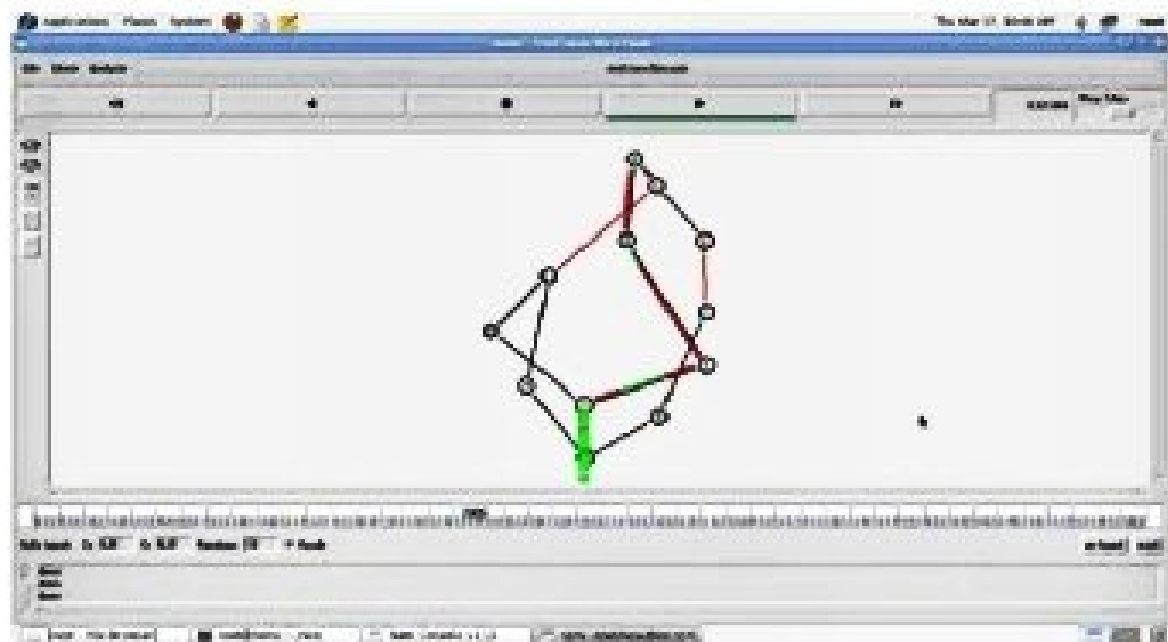
```
set ns [new Simulator]
set nr [open thro.tr w]
$ns trace-all $nr
set nf [open thro.nam w]
$ns namtrace-all $nf
```

```

proc finish { } {
global ns nr nf
$ns flush-trace
close $nf
close $nr
exec nam thro.nam &
exit 0
}for { set i 0 } { $i < 12 } { incr i 1 } {
set n($i) [$ns node]
for {set i 0} {$i < 8} {incr i} {
$ns duplex-link $n($i) $n([expr $i+1]) 1Mb 10ms DropTail }
$ns duplex-link $n(0) $n(8) 1Mb 10ms DropTail
$ns duplex-link $n(1) $n(10) 1Mb 10ms DropTail
$ns duplex-link $n(0) $n(9) 1Mb 10ms DropTail
$ns duplex-link $n(9) $n(11) 1Mb 10ms DropTail
$ns duplex-link $n(10) $n(11) 1Mb 10ms DropTail
$ns duplex-link $n(11) $n(5) 1Mb 10ms DropTail
set udp0 [new Agent/UDP]
$ns attach-agent $n(0) $udp0
set cbr0 [new Application/Traffic/CBR]
$scr0 set packetSize_ 500
$scr0 set interval_ 0.005
$scr0 attach-agent $udp0
set null0 [new Agent/Null]
$ns attach-agent $n(5) $null0
$ns connect $udp0 $null0
set udp1 [new Agent/UDP]
$ns attach-agent $n(1) $udp1
set cbr1 [new Application/Traffic/CBR]
$scr1 set packetSize_ 500
$scr1 set interval_ 0.005
$scr1 attach-agent $udp1
set null1 [new Agent/Null]
$ns attach-agent $n(5) $null1
$ns connect $udp1 $null1
$ns rproto LS
$ns rtmodel-at 10.0 down $n(11) $n(5)
$ns rtmodel-at 15.0 down $n(7) $n(6)
$ns rtmodel-at 30.0 up $n(11) $n(5)
$ns rtmodel-at 20.0 up $n(7) $n(6)
$udp0 set fid_ 1
$udp1 set fid_ 2
$ns color 1 Red
$ns color 2 Green
$ns at 1.0 "$scr0 start"
$ns at 2.0 "$scr1 start"
$ns at 45 "finish"
$ns run

```

OUTPUT:



NAME OF FACULTY: SANJANA SHARMA

SIGNATURE:

DATE: